MATH4511 Quantitative Methods for Fixed Income Securities
2026-2027 Fall Semester
Assignment 2

Submission deadline of Assignment 2: 11:59p.m. of 2nd Nov, 2026 (Mon)

Instruction: Please complete all required problems. Full details (including description of methods used and explanation, key formula and theorem used and final answer) must be shown **clearly** to receive full credits. Marks can be deducted for incomplete solution or unclear solution. Additional score will be given for well-written assignment.

Please submit your completed work via the submission system in canvas before the deadline. Late assignment will not be accepted.

Your submission must be (1) hand-written (you may write on the ipad if you wish), (2) in a single pdf. file (other file formats will not be accepted) and (3) contain your full name and student ID on the first page of the assignment.

Problem 1 (Required, 40 marks)

We consider an on-going interest rate swap. You are given that

- The interest rate swap has remaining life 15 months.
- The exchanges are made semi-annually in this swap and the next exchange will be made 3 months after today.
- The interest rate of floating rate payment is 6-month spot rate is $R_{\left[\frac{i-1}{2},\frac{i}{2}\right]}$ and the interest rate of fixed rate payment is 3.6%. The 6-month spot rate at the last exchange day is 3.7%.
- The spot rates of various maturities are given as follows:

	3 months	6 months	9 months	12 months	15 months
Spot rate	3.4%	3.5%	3.6%	3.7%	3.8%

- All above rates are quoted as annual nominal interest rate compounded semi-annually.

Find the current price of the swap using

(a) Original formula

(b) Pricing formula of floating rate bond and fixed rate bond.

(c) Forward rate

(d) Swap rate of the same swap quoted at current time.
(☺Hint: To determine the swap rate, you need to consider a swap with same payment schedule initialized at current time. Calculate the swap rate of this swap).

Problem 2 (Required, 30 marks)

Company A would like to borrow US dollars for 5 years and company B would like to borrow Canadian dollars for 5 years. The following table shows the borrowing rates offered by the banks in their home country.

	USD dollar	Canadian dollar
Company A	5.6%	5.0%
Company B	6.1%	6.8%

(*Note: The rates are quoted as annual effective interest rate)

(*Note: The companies repay the interest payment annually and repay the loan principal at the end of 5th year.)

(a) A financial institution would like to design a currency swap (with annual payments) that allows it to earn a profit of 0.3% in each exchange date and are equally attractive to two companies. Design a currency swap, if possible. Please explain your answer.
(*Note: In particular, please provide your explanation on how do you design the cashflows in the currency swap.)

(b) Suppose that the term structure of two currencies are flat and the annual effective interest rate of USD dollar and Canadian dollars are 5% and 4.5% respectively. The current exchange rate is 1 Canadian dollar = 0.85 USD. Suppose that the notional principal of USD payment and Canadian payment are chosen to be $N_{USD} = 85000$ and $N_{Cad} = 100000$ respectively. Calculate the value of the currency swap owned by Company B, in USD currency.

Problem 3 (Required, 30 marks)

We consider a 5-year interest rate swap signed today as follows:

- The notional principle is $N = \$10000$.
- The exchanges are made semi-annually and the next exchange will be made 6 months after today.
- The floating rate payment is 6-months spot rate applied over the last payment period (i.e. $R_{\left[\frac{i-1}{2},\frac{i}{2}\right]}$) and the interest rate of fixed rate payment is K (which is quoted as annual nominal interest rate compounded semi-annually.)

You are given that the annual nominal yield rate (compounded semi-annually) of a 5-year coupon bond with annual coupon rate 4% payable semi-annually is 3.9%. The face value of the bond is assumed to be $10000.

What can you say about the swap rate K of this interest rate swap? Choose the best answer below and provide full justification to your answer.

(A) Higher than 4%
(B) Equal to 4%
(C) Lower than 4%

Bonus Problem (Optional, 30 marks)

Companies A and B have been offered the following borrowing rates on a 5-year loan with loan principal $2,000,000.

	Fixed rate	Floating rate
Company A	5.0%	1-year spot rate + 0.1%
Company B	6.4%	1-year spot rate + 0.6%

(*Note 1: In other words, two companies can borrow $2,000,000 at either fixed interest rate or floating interest rate.)

(*Note 2: All rates are quoted as annual effective interest rate.)

(*Note 3: We assume that the borrower pays the interest payment (no principal repaid) annually during the first 4 years and repay the interest and loan principal at the end of 5th year)

(a) Company A would like to borrow money at a floating-rate and company B would like to borrow at fixed rate. Suppose that you are working in a financial institution, design an interest rate swap (with annual exchange) that will allow the institution to earn a net profit of 0.3% of the notional principal every year and is equally attractive to two companies. Provide full justification to your answer.

(☺Hint: The idea is similar to that of the foreign currency swap)

(b) Suppose that the current term structure is flat and the annual effective interest rate is 5%, calculate the value of the interest rate swap owned by Company A.

(c) Suppose that the borrowing rates offered to two companies are as follows:

	Fixed rate	Floating rate
Company A	5.5%	1-year spot rate + 0.1%
Company B	6.4%	1-year spot rate + 0.9%

Design another swap, if possible, that can achieve the same objective as in (a).